

A Bioinspired Spiking Attention Neuromorphic Accelerator for Low-Power Visual Prosthesis

Shiqi Zhao^{ID}, *Member, IEEE*, Qi Wang^{ID}, Zhongcheng Shu, Xinrui Yang^{ID}, Miao Fang^{ID},
Jie Yang^{ID}, *Senior Member, IEEE*, and Mohamad Sawan^{ID}, *Life Fellow, IEEE*

Abstract—Retinal degenerative diseases have led to moderate-to-severe vision loss in over 230 million people globally. Visual prostheses aim to restore partial perception by converting external visual stimuli into neural-response patterns that can be delivered to the remaining visual pathway. Spiking neural models are particularly suitable for this task because they represent visual information in an event-driven temporal form that is compatible with neural signaling. However, existing approaches often rely on biologically oversimplified hierarchies and insufficient modeling of retinal-layer interactions and retina-to-cortex dynamics, limiting neural-response fidelity under complex visual stimuli. Meanwhile, implantable visual prostheses impose strict power and area constraints, making algorithm-hardware codesign essential. This article presents a bioinspired neuromorphic visual prosthesis framework that tightly couples spiking neural modeling with specialized hardware architecture. The proposed model emulates the hierarchical organization of the human retina through multiscale spiking convolutional layers that simulate photoreceptors, bipolar cells, and ganglion cells. Within this retinal hierarchy, a spiking attention mechanism is incorporated to model bottom-up visual dynamics and adaptive information modulation, enabling efficient spatiotemporal feature selection consistent with biological visual processing. A spiking linear readout layer is further employed to predict primary visual cortex (V1) responses to natural images. On the hardware side, we design a neuromorphic accelerator optimized for visual prosthesis tasks. The architecture integrates a multiscale sparsity exploitation strategy with a multistage pipelined parallelism scheme to maximize energy efficiency and minimize area overhead. From an instrumentation perspective, the framework integrates event-driven visual sensing, spike-domain signal processing, and quantitative neural-response evaluation under implantable hardware constraints. Under 8-bit quantization, the proposed framework achieves a Pearson correlation of 0.713 for V1 response prediction. ASIC postsynthesis estimation under the TSMC 40-nm CMOS process reports a power consumption of

199.10 mW and a chip area of 7.89 mm². Under ZCU106 FPGA deployment, the architecture achieves a throughput of 79.7 GOP/s and a peak energy efficiency of 259.2 GOPS/W.

Index Terms—Bottom-up attention, neuromorphic accelerator, spiking attention, visual prosthesis.

I. INTRODUCTION

CURRENTLY, due to retinal degenerative diseases, up to 230 million people worldwide suffer from moderate to severe vision loss [1]. Visual prosthesis chips are currently the only technology capable of restoring visual function in patients with such conditions [2]. These chips typically consist of sensors, processing units, and electrical stimulators, and work by mimicking the human visual perception pathway—bypassing degenerated cellular layers to stimulate the remaining visual neurons.

Existing research on visual prostheses has primarily focused on accurately predicting the responses of target cells [3], [4]. These efforts often employ artificial intelligence algorithms that simulate the functions of the visual perception pathway using various network architectures. Although current methods can partially predict neural responses, most rely on simplified neural hierarchies with limited biological detail [5]. In particular, they lack fine-grained modeling of the interactions between retinal cell layers and fail to simulate the dynamic response of the retina-cortex pathway. Such simplified modeling cannot be readily generalized to complex visual stimuli. As a result, it fails to capture the true complexity of the visual system and limits the performance of existing visual prostheses. Moreover, visual prosthesis chips typically need to be implanted and fixed onto the human retina [6]. Therefore, their design must meet strict requirements for long-term stable power supply and compact size to accommodate the limited space and prolonged operation within the human body. However, most current studies separate algorithm design from hardware design, resulting in suboptimal tradeoffs in processing efficiency, power consumption, and chip area.

To address these bottlenecks, we propose a bioinspired neuromorphic visual prosthesis framework based on codesign of algorithm and hardware. Inspired by the human retina's working mechanisms, this framework models the interactions between different retinal cell layers in detail. In particular, external scene information is encoded using spike-based representations and processed through spike neurons. We implement multiscale spiking convolution layers to emulate the response characteristics of photoreceptors, bipolar cells, and ganglion cells. Furthermore, drawing from the bottom-up attention mechanism of the human visual

Received 25 February 2026; revised 24 June 2026; accepted 1 July 2026. Date of publication 27 July 2026; date of current version 4 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62404037 and Grant 623B2085, in part by Hebei Natural Science Foundation under Grant F2024501017, in part by the Fundamental Research Funds for the Central Universities under Grant Z2026-057, and in part by Shijiazhuang—Northeastern University Science and Technology Cooperation Special Project under Grant NEUS2025-01-003. The Associate Editor coordinating the review process was Dr. Benkuan Wang. (Corresponding author: Miao Fang.)

Shiqi Zhao, Qi Wang, Zhongcheng Shu, Xinrui Yang, and Miao Fang are with Northeastern University, Shenyang, Liaoning 110819, China, also with Hebei Key Laboratory of Marine Perception Network and Data Processing, Northeastern University at Qinhuangdao, Qinhuangdao, Hebei 066004, China, and also with Shijiazhuang Innovation Research Institute, Northeastern University, Shijiazhuang 052165, China (e-mail: fangmiao@neuq.edu.cn).

Jie Yang and Mohamad Sawan are with the Center of Excellence in Biomedical Research on Advanced Integrated-on-Chips Neurotechnologies (CenBRAIN Neurotech), School of Engineering, Westlake University, Hangzhou 310024, China, and also with the Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou 310024, China.

Digital Object Identifier 10.1109/TIM.2026.3712917

1557-9662 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: Westlake University. Downloaded on August 10, 2026 at 08:47:48 UTC from IEEE Xplore. Restrictions apply.

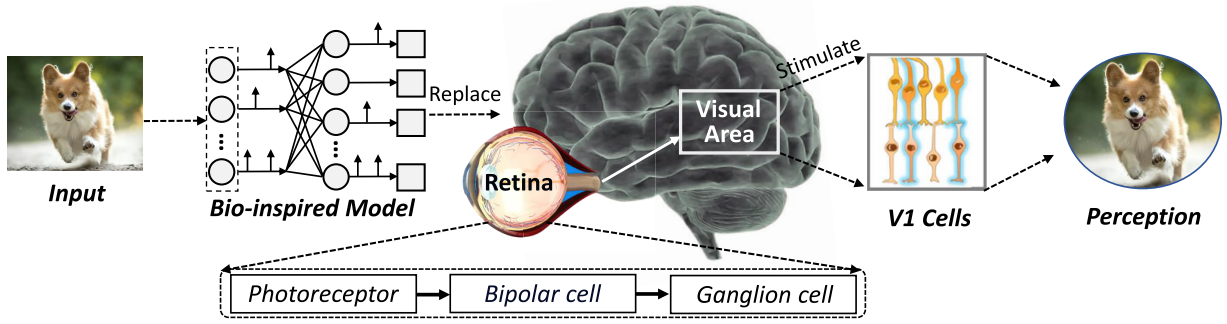


Fig. 1. Working principle of visual prosthesis. It encodes external images into a model-interpretable format using sensors. It then employs a bioinspired model to simulate the functions of the damaged retina and visual pathways, generating predicted responses of visual areas such as V1 cells. This process enables the brain to construct a perceptual map of the external images.

system, we design a novel spiking attention module to accurately reflect the dynamic response of the retina-to-cortex pathway. Finally, we introduce a spiking linear layer to predict responses of primary visual cortex (V1) neurons, validating the system's ability to respond to real image stimuli.

Based on this algorithmic framework, we design a novel neuromorphic accelerator architecture for visual prosthesis applications. By analyzing the data characteristics of each module, we propose a multiside sparsity exploitation strategy, which maximizes sparsity utilization while minimizing area overhead through reconfigurability. In addition, we employ a multistage pipelined parallel processing scheme to optimize both intermodule and intramodule parallelism, further improving energy efficiency. The main contributions of this work are summarized as follows.

- 1) *Bioinspired Visual Perception Modeling*: We design three spiking convolution layers to emulate the behavior of photoreceptors, bipolar cells, and ganglion cells. Inspired by the bottom-up attention pathway of the human visual system, we innovatively introduce a spiking attention mechanism to enhance the system's response to realistic image stimuli. A spiking linear layer is used to predict responses of V1 neurons.
- 2) *Neuromorphic Architecture for Visual Prosthesis*: We present the first neuromorphic accelerator tailored for visual prosthesis, featuring a pipelined architecture that maximizes temporal data reuse. For retinal cells, spiking attention, and V1 output layers, innerproduct and outerproduct sparsity strategies are employed to enhance performance and efficiency.
- 3) *Experimental Results*: Under 8-bit quantization, our system achieves a Pearson correlation coefficient of 0.713 for V1 neuron response prediction, surpassing existing methods. Moreover, the proposed processor achieves comparable power consumption and throughput to prior spiking neural network (SNN) accelerators, while improving energy efficiency.

II. BACKGROUND AND MOTIVATION

A. Visual Prosthesis

The normal visual pathway involves visual scene information entering the eye, being processed by retinal neurons, and then being output as neuronal spikes, which are sent to the cortex for further computation. For individuals with

partial visual impairment, however, certain retinal neurons cannot autonomously complete this neural pathway. Visual prosthetics are precision medical devices designed for specific visual impairments, aiming to manipulate neural spike signals in a closed-loop manner. They bypass damaged retinal areas to directly stimulate intact V1 cells while receiving external environmental stimuli. The basic working principle of visual prosthesis is shown in Fig. 1.

Over the past decade, several visual prosthetics have been developed and even implanted in the retinas of some patients. For instance, second sight's Argus II received FDA approval in the United States in 2013 for restoring vision in adults with retinitis pigmentosa [7]. Alpha AMS, a subretinal prosthetic, functions by replacing photoreceptors in both space and functionality through implantation in the subretinal space [8]. Although these devices have improved the visual localization abilities of patients to some extent, the limitations in processing power and resolution have prevented fully satisfactory outcomes.

To overcome these limitations, recent research on visual prosthetics has incorporated artificial intelligence algorithms, primarily focused on accurately predicting target cell responses [5]. These efforts typically abstract the retinal cell mechanisms into multiple network modules, leveraging classical neural network models like CNNs and RNNs [4]. For example, Yan et al. [3] introduce a three-layer convolutional neural network to link visual stimuli with retinal perception. Inspired by human retinal structure, McIntosh et al. [9] add recurrent connections to CNNs to enhance model fitting, improving the accuracy of neuronal response prediction. However, to truly serve blind individuals, algorithms for visual prosthetics need deployment on appropriate hardware systems, requiring models that can simulate retinal mechanisms while maintaining hardware-compatible complexity [9]. Ho et al. [10] propose a CNN model based on the PRIMA system, achieving partial hardware deployment for visual prosthetics, though the recognition speed remains slower than natural visual processing.

To reduce the information processing redundancy inherent in conventional CMOS sensors, recent studies have explored bioinspired dynamic vision sensors (DVSs). These sensors respond only to scene changes, enabling low-power operation and naturally aligning with the spike-based coding of retinal neurons [11]. Wang et al. [12] propose an SNN-based method to predict the responses of V1 cells, achieving a Pearson correlation coefficient of 0.65 while reducing theoretical

energy consumption by over 15 times. While spike-processing research in visual prosthetics has made significant progress, most existing work remains at the algorithmic level, limiting the full demonstration of spike-based processing advantages.

B. Neuromorphic Accelerators

Spike-based visual prosthetics require hardware–software codesign with neuromorphic accelerators to fully leverage their event-driven, low-power, and bioinspired advantages. Existing neuromorphic accelerators, which typically utilize common SNN algorithms and nonvon Neumann architecture, are known for their event-driven characteristics. This approach allows neuromorphic accelerators, such as IBM’s TrueNorth and Intel’s Loihi [13], [14], to achieve ultralow power consumption. In particular, they integrate neuron and processing modules, address neural network computation as an addressing problem, and utilize input spike sparsity to determine which neurons’ membrane potentials need to be updated. This sparse, spike-driven computation mechanism enables significantly lower energy consumption compared to ANN accelerators.

In recent years, numerous studies have introduced novel neuromorphic accelerators. Unlike ANN accelerators, existing SNN accelerators face challenges with energy efficiency due to multitime-step requirements. Consequently, most neuromorphic accelerator research focuses on further enhancing energy efficiency. Some approaches optimize for the spike sequence characteristics to improve efficiency. For instance, SpinalFlow optimizes the computation sequence to compress neuron outputs, leveraging the sparsity of spike output sequences to reduce storage overhead [15]. SATO redefines the SNN dataflow, removing dependencies across time steps [16]. In addition, recognizing the distinct applications of ANN and SNN accelerators, some researchers have proposed specialized heterogeneous architectures combining the strengths of both. C-Transformer, for instance, is a specialized heterogeneous architecture for DNN-Transformer/spiking-Transformer tailored to large language models [17]. Fang et al. [18] designed an efficient heterogeneous architecture by addressing memory redundancy and scheduling issues in spiking attention.

However, current neuromorphic accelerators struggle to meet the demands of visual prosthetics. First, due to application limitations, they lack specialized operator support for the visual pathway, limiting optimal power consumption and efficiency. Second, in visual prosthetics, the dynamic nature of operators modeling retinal cell functionality makes heterogeneous architectures less suitable. Finally, we observe that, in addition to sparsity in spike sequences, there is also sparsity in weights and other operands that has not been fully explored in existing works, constraining energy efficiency. Therefore, we propose a neuromorphic accelerator specifically for visual prosthetics, integrating customized operator optimizations for the model and inner/outer product-based designs to address multisided sparsity.

Although existing neuromorphic accelerators have demonstrated significant progress in energy-efficient SNN inference, most architectures are optimized for general-purpose workloads such as image classification or event-based recognition.

On the contrary, visual prosthesis systems require biologically interpretable neural response prediction under strict implantable constraints. This difference introduces several additional design challenges, including continuous spike-stream processing, stable low-power operation under varying visual complexity, biologically inspired temporal dynamics, and efficient support for spike-transformer operators. Consequently, architectures designed for conventional SNN inference cannot be directly applied to implantable visual prosthesis systems without substantial modification.

III. PROPOSED VISUAL PROSTHESIS ALGORITHM

Current visual prosthesis algorithms face significant challenges in simulating the complex functions of the human retina, particularly in accurately modeling intercellular interactions across retinal layers and reproducing the dynamic responses of the retina-to-cortex visual pathway. These limitations stem from the fact that most existing studies focus primarily on accurately predicting the responses of target cells, often relying on simplified neural hierarchies and limited biological details. As a result, their precision is constrained when dealing with complex visual stimuli, making it difficult to faithfully replicate the intricacies of the visual system. This chapter presents the algorithmic component of the proposed bioinspired neuromorphic visual prosthesis framework.

A. Bioinspired Architecture

The retina comprises three main components: photoreceptors, bipolar cells, and ganglion cells. Photoreceptors, located in the outermost layer of the retina, capture light signals and convert them into neural spike signals. Bipolar cells receive these signals and transmit them to ganglion cells, which consolidate the information and send it to the brain’s V1 (e.g., V1 area) via the optic nerve for further processing. To simulate the neural spike transmission within the retina and the receptive fields of natural retinal cells, a three-layer spiking convolutional network was designed to model the processing behavior of the three retinal cell layers, as shown in Fig. 2. In addition, to meet the power and area requirements for implantable visual prosthetics, each of the three retinal cell layers employs depthwise separable convolution (DSC), reducing computational and parameter demands. The readout layer effectively minimizes correlation between predicted responses of different neurons, enhancing prediction accuracy. Finally, a spike-counting layer was used to accumulate spikes generated during the prediction process.

The retinal cell layers consist of three primary components: photoreceptors, bipolar cells, and ganglion cells, which work in concert to process light signals and convert them into neural information. Photoreceptors, located in the outermost layer of the retina, are responsible for capturing incoming light. These include two types of photoreceptive cells: rods and cones. Rod cells are highly sensitive to low-light conditions but do not detect color, while cone cells are responsible for color perception by responding to different wavelengths of light. The photoreceptors convert captured light into electrochemical signals and transmit them via synapses to downstream bipolar cells. As intermediaries, bipolar cells receive input from

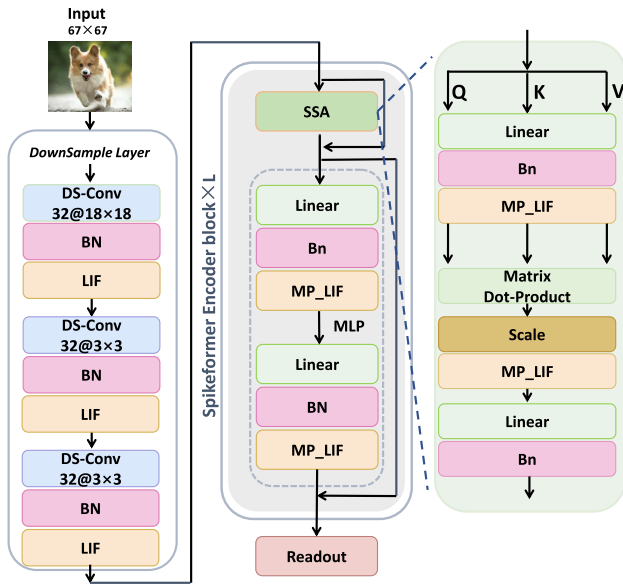


Fig. 2. Proposed algorithm model is outlined. First, three DSC layers are used to simulate the functions of the retinal photoreceptors, bipolar cells, and ganglion cells. This is followed by a spike-based self-attention layer to model the bottom-up attention mechanism of the human visual pathway. The readout layer is implemented through a linear layer connected to 166 MPLIF neurons, which predicts the responses of 166 V1 neurons.

photoreceptors and relay it to ganglion cells. Ganglion cells integrate these signals and send the processed information to the brain's V1 (e.g., the V1 area) through their axons, which collectively form the optic nerve.

To simulate the process of neural spike transmission and the receptive field characteristics within the retina, we designed a three-layer multiscale spiking convolutional network architecture inspired by the functional specialization of retinal cells. First, the process by which photoreceptors capture light and convert it into electrochemical signals is modeled in the first layer of the network. The convolutional operations in this layer simulate the responses of rod and cone cells to light stimuli, ensuring that the network can sensitively capture the primary spiking information of external visual input. Next, the role of bipolar cells is represented in the second convolutional layer, which integrates and preliminarily processes signals from the photoreceptors, similar to signal enhancement and relay in the biological retina. Finally, the third convolutional layer simulates ganglion cells, integrating signals and transmitting them through spiking activity to higher levels, analogous to how processed retinal information is conveyed to the brain.

This layer-by-layer simulation design closely mirrors the division of labor among retinal cells in biological visual information processing, ensuring that each convolutional layer in the neural network emulates the functional role of its biological counterpart. The weight parameters of these three layers are shown in the figure. Each of the first three convolutional layers requires 80 000 integrate-and-fire (IF) neurons. These biologically inspired IF neurons, with multiple spiking rates, provide nonlinear support for the convolutional outputs. Moreover, to meet the stringent power consumption and size constraints of implantable visual prostheses, we adopt a DSC structure in each layer. This design significantly reduces computational complexity while maintaining

processing capability across layers, thus achieving both biologically faithful modeling and high efficiency. Importantly, the mapping between the three spiking convolutional layers and the retinal cell stages should be interpreted as a biologically motivated functional abstraction rather than a one-to-one biophysical replica. The biological motivations include the hierarchical signal-transmission order, local receptive-field processing, spike-based communication, and temporal integration of IF neurons. The engineering simplifications include trainable convolution kernels and DSC, which approximate functional responses while omitting detailed cell-type dynamics, synaptic connectivity, and feedback circuitry.

B. Spiking Bottom-to-Top Attention

The proposed biomimetic visual prosthesis algorithm not only replicates the functional behavior of retinal cellular layers but also emulates the retina's core physiological mechanisms. Far from being a passive entry point for visual information, the retina actively participates in responding to external light stimuli in coordination with the cerebral cortex, forming a highly adaptive perceptual system attuned to environmental dynamics. In particular, the retina interacts with the brain through two primary feedback loops that constitute the visual attention mechanism: the top-down and bottom-up attention pathways. The top-down pathway is closely associated with goal-directed processing in the cortex. It regulates the retinal responses based on the cortex's selective attention to specific visual targets, enabling the enhancement or suppression of information processing in accordance with task demands. On the contrary, the bottom-up pathway is driven directly by external visual stimuli. Based on the detected visual features, the retina autonomously allocates attention, involving the dynamic regulation of its hierarchical structure and the interactions among photoreceptors, bipolar cells, and ganglion cells. These self-regulatory adjustments support efficient perceptual encoding and rapid environmental awareness. However, in individuals with visual impairments, the bottom-up pathway is often disrupted, leading to loss of stimulus-driven visual attention. Therefore, the proposed visual prosthesis model is designed to simulate this physiological phenomenon—particularly the bottom-up attention mechanism—with the goal of restoring the retina's automatic regulatory capacity in visual information perception.

As illustrated in Fig. 2, to simulate the bottom-up attention pathway in the human retina, we introduce a spiking self-attention (SSA) layer. In SSA, the query, key, and value matrices are all represented in spiking form (binary values 0 and 1). This design is inspired by the discrete response characteristics of retinal neurons and the integrative mechanism of the bottom-up attention pathway. In biological systems, external visual stimuli directly activate photoreceptors (e.g., cone and rod cells) in the retina. The intensity and pattern of these stimuli influence the spiking frequency and temporal firing patterns of retinal neurons, which in turn determine the prioritization of information processing in the retina. This fusion of discrete neural responses and attention mechanisms reflects how the nervous system reacts rapidly to environmental changes without conscious intervention. In

the model implementation, spiking encoding mimics neuronal spike trains to reproduce this biological mechanism. It effectively captures the dynamic characteristics of visual stimuli and enables automatic attention modulation.

Unlike conventional self-attention methods such as vanilla self-attention (VSA), which rely on a softmax layer to compute attention weights, SSA leverages the inherent nonnegativity of the spiking matrices, eliminating the need for softmax. This innovation not only improves computational efficiency but also aligns more closely with the way neural signals are processed in biological systems, particularly the communication between the retina and the cortex. Moreover, SSA dynamically adjusts its computation sequence based on the relationship between the input sequence length N and the head dimension d : if $N < d$, QK^T is computed first, with a complexity of $O(N^2d)$; otherwise K^TV is computed first, with a complexity of $O(Nd^2)$. This adaptive computation mechanism not only enhances model performance but also reflects the flexibility observed in bottom-up visual processing. By simulating the dynamic response of the retina–cortex pathway, the SSA layer more accurately reproduces the firing patterns of retinal ganglion cells in response to visual stimuli. In addition, the rest of the model incorporates spiking-form multilayer perceptrons (MLPs), further reinforcing the biologically inspired simulation of retinal processing mechanisms.

C. Spiking Readout Module

To ensure that the predictive responses align closely with actual recordings of neurons in the primary V1 across both temporal and spatial domains, we introduce a nonlinear readout layer to optimize feature decoding and output. In the temporal domain, we introduce tunable parameters to finely adjust the firing patterns of individual neurons, enabling their responses at different time points to accurately fit the recorded dynamics of real V1 neurons. Biological studies have shown that neurons in the V1 region exhibit significant temporal variability in their responses to visual stimuli, with spike train patterns displaying complex dynamics that evolve with the input stimuli. To emulate this behavior, the model's time-response parameters are calibrated such that its output spike sequences closely mirror the temporal structure observed in biological recordings. In the spatial domain, we implement a tailored loss function to regulate interneuronal correlations, aiming to reduce redundancy and promote functional independence among neurons. This approach is inspired by the spatial decorrelation and functional specialization found in biological neural circuits, where individual neurons tend to process localized input features with minimal reliance on collective synchronization. The decorrelation mechanism ensures that the model maintains high spatial resolution and heterogeneity in its output representations. Through this dual-domain design, the model exhibits enhanced adaptability to the diverse temporal and spatial response patterns of V1 neurons. Such expressiveness is particularly vital for visual prosthesis systems, which must replicate the spatial independence and diversity of cortical neurons to enable accurate reconstruction of complex visual scenes.

Each spiking readout module contains a spiking MLP layer and a batch normalization layer. In addition, similar to the

spiking attention layer, we incorporate a membrane potential leak-integrate-and-fire neuron (MPLIF) model to enhance computational efficiency and diversity. Unlike traditional LIF neurons, the MPLIF neuron omits the comparison step and outputs the neuron's membrane potential, with its dynamic properties represented as follows:

$$V_t = \left(1 - \frac{1}{\tau}\right) V_{t-1} + \left(\frac{1}{\tau}\right) X_t \quad (1)$$

where the balance between forgetting the previous potential V_{t-1} and incorporating new input X_t is controlled by the time constant τ , enabling functionality similar to recurrent connections. To ensure that the predicted responses of the proposed model closely resemble recorded values from V1 cells in both temporal and spatial domains, we use a nonlinear readout layer. In the temporal domain, adjustable parameters allow each neuron to accurately fit recorded values across different time points. In the spatial domain, the loss function minimizes correlations among neurons, promoting flexibility for each neuron to independently fit the recorded data.

IV. HARDWARE ARCHITECTURE

A. Overview

At the hardware level, we propose a pipelined visual prosthetic architecture with both intramodule and intermodule pipelining. This design aims to maximize computational resource utilization and data reuse across spike time steps. The overall architecture is shown in Fig. 3, containing four main parts: the decoder, retina cell module, spiking attention module, and V1 readout module. External data are captured through a DVS camera and then loaded onto the chip via a direct memory access (DMA) interface, which also loads pretrained parameters and input data. The decoder serializes the input data for processing. The retina cell module simulates photoreceptors, bipolar cells, and ganglion cells, each primarily built with spiking convolution processing elements (SConvPEs). The weights and membrane potentials of the neurons are stored in dedicated SRAM within each module. The computations for each layer are accumulated in their respective cell modules to form the final structure. The attention module consists mainly of spiking attention processing elements (SAPEs) for performing attention operations. Skipping logic is implemented to bypass entire rows or columns with all-zero values in attention calculations, enhancing efficiency. Partial sums are accumulated through a merged adder tree. The main processing element in the V1 readout module is the spiking linear processing element (SLPE), responsible for spiking MLP computations. A residual module handles residual connections in the model. The output SRAM stores the cumulative membrane potentials of output neurons across multiple time steps, generating the final output.

B. Multiside Sparsity Exploitation

To further improve energy efficiency, we propose a multiside sparsity exploitation strategy tailored to different operators to leverage the sparsity within the model, as shown in Fig. 4. In the retina cell module, the core operation is the spiking convolution. First, the filter is transformed into a spike-like mask

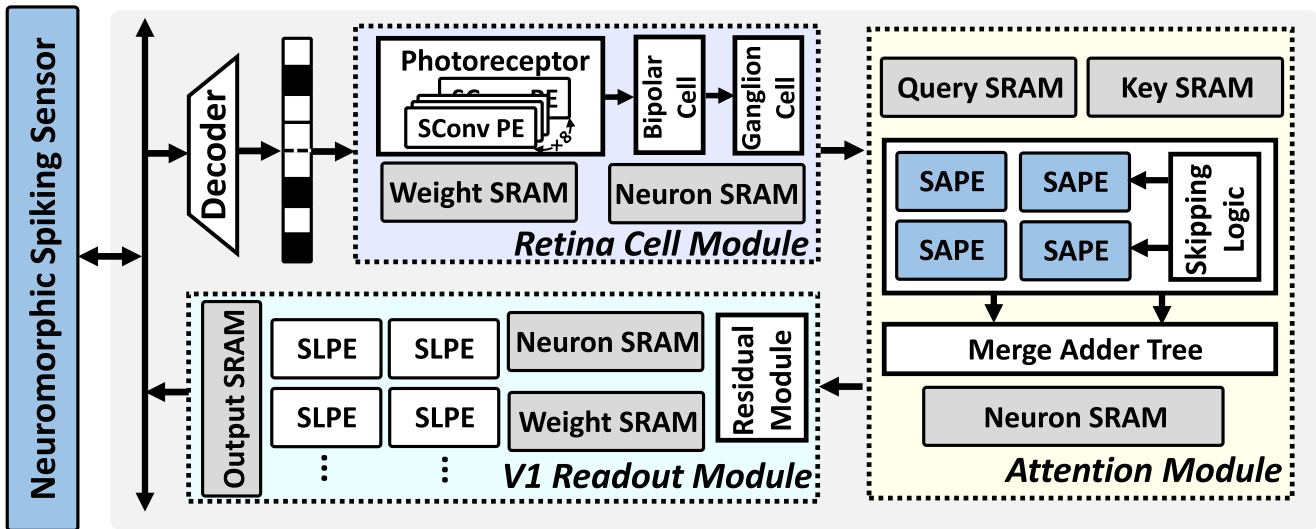


Fig. 3. Overall architecture of the proposed visual prosthesis chip. The external input is generated by a neuromorphic DVS sensor and serialized through a decoder. The processed spike data is then fed into the retina cell module, which includes photoreceptors, bipolar cells, and ganglion cells. Each module contains several spiking convolution PEs. The output from the ganglion cells is sent to the attention module, where the core computational unit is the spiking attention PE, which supports dynamic zero-vector skipping based on matrix outer products. The results are then passed to an adder tree for merging, and the output of the attention module is forwarded to the V1 readout module. The core computational unit in the V1 readout module is the spiking linear PE, which can perform sparse spiking matrix multiplication. The residual module supports the computation of residual connections in the model. Finally, the membrane potential output of 166 V1 neurons is generated.

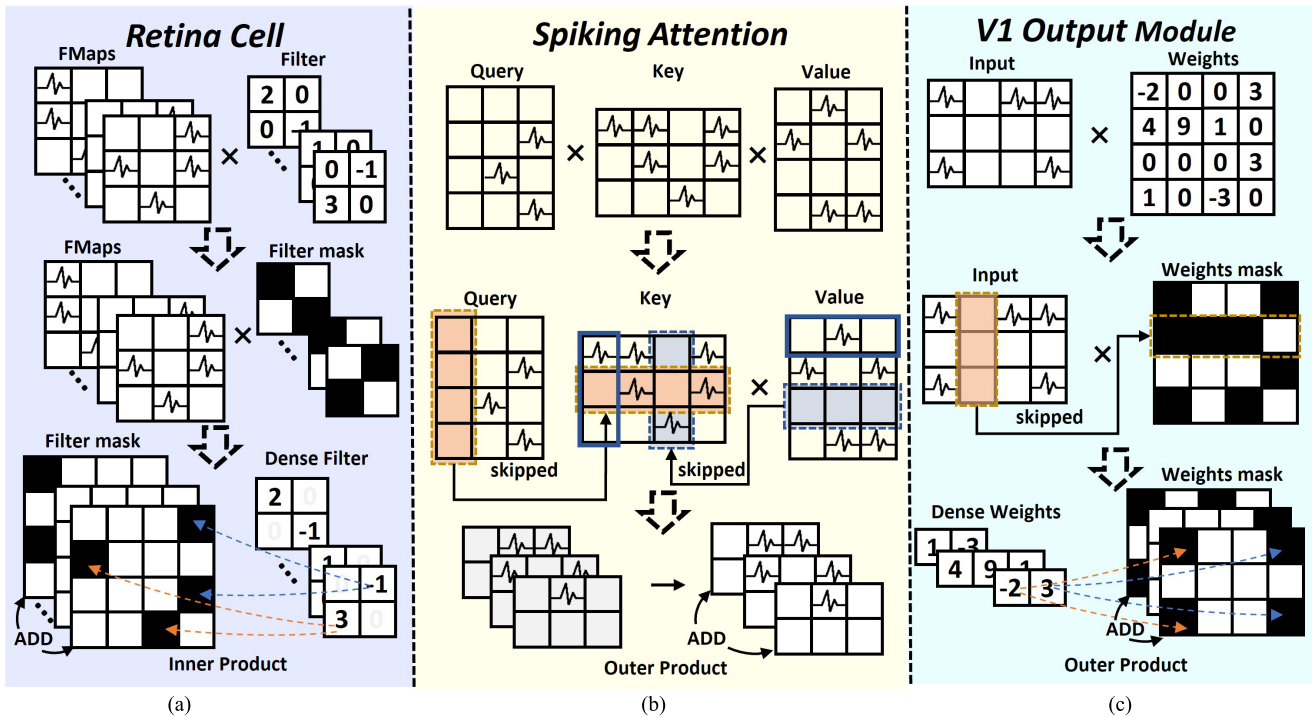


Fig. 4. Multiside sparsity exploitation. (a) For the retina cell, where spiking convolution is the primary operation, the nonzero values of the filter are first located and used as a bitmap mask. This mask is then used to perform an inner product operation with the spiking input feature map, leveraging the inherent data reuse of the filter in the convolution operation. After obtaining the mask result, the nonzero values are mapped to their corresponding positions, and the feature map dimensions are accumulated. (b) For the spiking attention module, the computation order is determined based on the sizes of N and d . Sparse outer product operations are then performed on the two precomputed spike matrices, with calculations skipped according to the nonzero rows/columns of the other matrix. The partial outer product results are accumulated to obtain the final output. (c) In the V1 readout module, the core operation is spiking linearly. First, the nonzero weight matrix is processed to obtain the bitmap mask matrix. A sparse outer product matrix multiplication is then performed, similar to the spiking attention module, to compute the partial bitmap results. After mapping the nonzero values to their corresponding positions, the results are accumulated to yield the final output.

matrix. The result of the inner product between this mask and the spike input is used as an enable signal to access the corresponding weight values. Only nonzero weights are retrieved, replacing the “1” values from the inner product, which

effectively skips storing and computing many zero-valued weights. Partial sums from different feature maps are then accumulated to form the final output, activating neurons that serve as inputs for the next layer.

In the spiking attention module, the query, key, and value are binary spike matrices (0,1). We propose a highly parallelized spike outer product operation that takes advantage of bidirectional sparsity. The dimensions of the operators are determined by sequence length N and head dimension d . In particular, for vectors like query, key, or value retrieved from SRAM, the skipping logic evaluates whether the respective rows or columns of the matrix are zero, allowing certain vectors to be bypassed. When performing outer product calculations in the SAPE, the spike-driven design further controls which registers are gated. The results of these computations are sent directly to an adder tree for independent merging and neuron firing.

In the V1 readout module, the core computation is the spiking MLP. We convert the weights into a mask matrix format, making the spiking MLP computation similar to the spiking attention operation. We adopt an outer product computation format with bidirectional sparsity, mapping nonzero values to the corresponding positions in the partial sums. Finally, these partial sums are accumulated to generate the V1 output predictions.

To enhance cross-time-step data reuse and accelerate the overall computation pipeline, we adopt a multistage pipelined parallel processing approach, as illustrated in Fig. 5. The proposed architecture mimics the structure of the human retina by constructing a multimodule/cell parallel processing pipeline. After the decoder serializes the input, the retina cell module begins processing the first spike train. When the second spike train arrives, data from both the first and second spike trains can be processed concurrently in the retina cell and attention modules. While the V1 output module processes the first spike train, the attention and retina cell modules simultaneously handle the second and third spike inputs. Each module also contains an internal parallel processing pipeline. Each spike train consists of multiple time steps, and the same neuron accumulates membrane potentials across time steps. This enables each time step's accumulated potential in neurons across modules to be reused directly in the next time step, reducing resource consumption by avoiding repeated SRAM reads.

Fig. 6 illustrates the detailed circuit design of three PEs. For convolution operations, the input data are first converted into a column data format via the im2row module and reads specific positions of the matrix based on the data address. Each column of data undergoes elementwise multiplication with the mask of the convolution kernel weights. Based on the operation results, the corresponding values of matrix B are selected and then passed to the accumulator for accumulation. The accumulated results are stored in registers and evaluated by a multiplexer to determine if they are zero. Finally, the results are fed into an adder tree for addition operations, completing the accumulation across multiple channels to generate the final convolution output. The input spike signal is selected from the input feature map or skip connection through a multiplexer. By performing a logical OR operation on all data, zero-detection is conducted on the rows and columns of matrices A and B. If the detection result indicates all zeros, the computation for that row/column is skipped; otherwise, an outer product operation is performed, followed by accumulation. The accumulated results are output to registers and evaluated by a multiplexer to check for zero

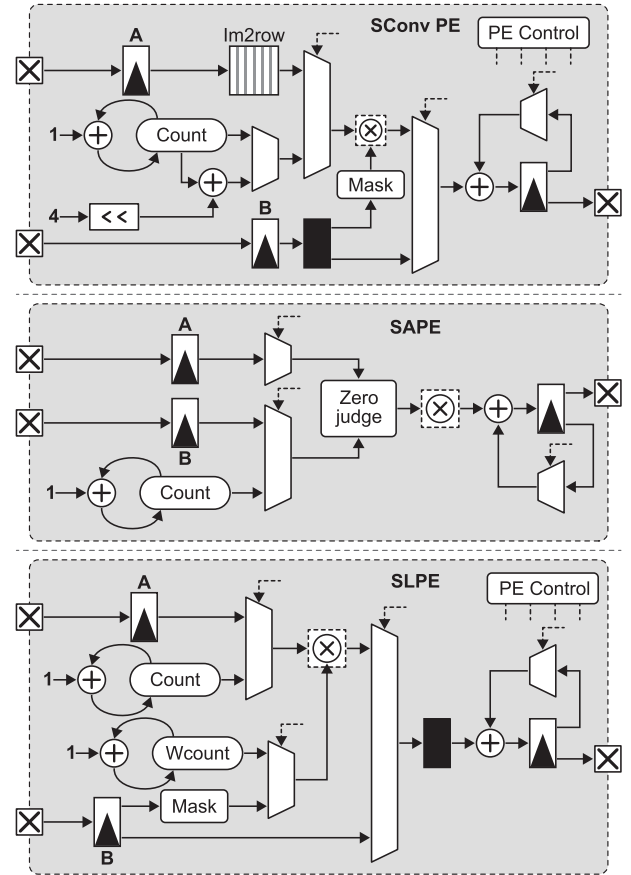


Fig. 5. Circuit architectures of the three processing elements in the proposed processor. SConv PE performs sparse spiking convolution with im2row and weight masks; SAPE skips zero rows/columns for outerproduct attention; SLPE executes sparse spiking linear accumulation using count/Wcount addressing.

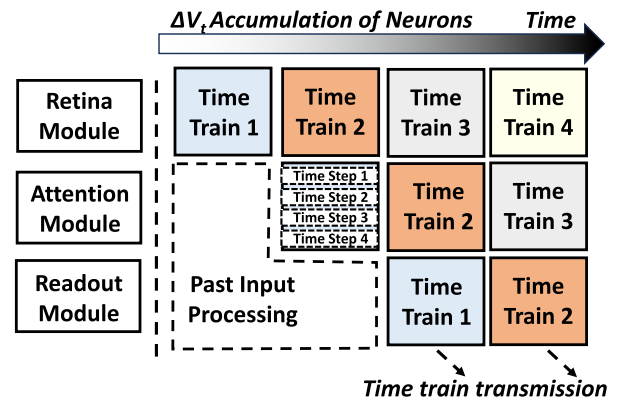


Fig. 6. Parallel computation is performed in a pipelined manner across modules, with each module operating on spiking groups. In addition, within each module, a pipeline is implemented based on the spiking time steps.

values. Subsequently, addition operations are performed to generate the result of the spike matrix multiplication. The input feature map is selected based on the address provided by the count, while the mask of the weight data is read according to the Wcount address. The data are first subjected to elementwise multiplication with the weight mask. Based on the resulting values, the corresponding entries of matrix B are selected and forwarded to the accumulator for aggregation.

The accumulated results are stored in registers, evaluated by a multiplexer to determine if they are zero, and then fed into an adder tree for addition operations, completing the accumulation across multiple channels to produce the final result.

C. Workflow

As shown in Algorithm 1, the data address generation mechanism for convolution operations is introduced using SConv PE as an example, aiming to achieve precise and dynamic data addressing to support computations for multilayer convolution operations. By coordinating a set of key parameters, including the state register, counter, image index, stride, and input/output data addresses, the mechanism ensures tight synchronization between data access and computation progress. Its core functionality is implemented through four modules. The data address generation module dynamically adjusts the output addresses based on the count flag under layer processing states. When `count_next` is true, the address is set to the current counter value to ensure synchronous updates; otherwise, the input address is used to maintain consistency, thereby optimizing memory access efficiency. The counter update module increments the counter during the computation state when the start signal is activated, precisely tracking the operation progress and avoiding redundant computations. The counter update decision module evaluates the `count_next` flag using a switch statement. Under layer processing states, it determines whether to advance address generation based on the index threshold, or in the `ST_SEND` state, based on the transmission signal, ensuring synchronization with computation or transmission requirements. The index update module computes the input address by concatenating count and image_index and sets the index value based on state changes or the countflag (7 for `ST_LAYER1`, 8 for `ST_LAYER2/ST_LAYER3`, or reset to 0 by default) to enable data traversal. If no state change occurs, the index is incremented to support continuous addressing. These modules collectively ensure that the PE can flexibly adapt to the requirements of different convolutional layers.

The stride parameter supports various convolution kernel sizes and sliding window strategies, enhancing versatility. By reducing memory access overhead and optimizing data flow management, the mechanism significantly improves computational efficiency.

Although the proposed architecture is evaluated against several representative FPGA-based SNN accelerators, we note that direct one-to-one comparison across different works is inherently challenging due to differences in application targets, neural network structures, FPGA platforms, clock frequencies, precision settings, and evaluation methodologies. Existing accelerators are often optimized for general-purpose SCNN or SFNN workloads, whereas the proposed design specifically targets biologically inspired visual prosthesis processing with spike-transformer operators and retinal-inspired dataflow characteristics. Therefore, the comparison results in this work are intended to provide relative efficiency and architectural trend references rather than absolute equivalence under identical deployment conditions. To improve transparency and repro-

Algorithm 1 Data Address Generation Mechanism for Convolution Operations

```

1: Initialization:
2:  $state\_reg \leftarrow ST\_IDLE, \quad count \leftarrow 0,$ 
    $image\_index[3:0] \leftarrow 0$ 
3:  $stride \leftarrow 1, \quad data\_addr[0] \leftarrow 0, \quad data\_addr[1] \leftarrow 0$ 
4:  $count\_next \leftarrow false, \quad send\_cnt \leftarrow 0$ 
5: for each state transition of  $state\_reg$  do ▷Cyclic update
   driven by FSM transitions
6:   Module 1: Data Address Generation
7:   if  $state\_reg = ST\_LAYER1 \vee state\_reg =$ 
      $ST\_LAYER2 \vee state\_reg = ST\_LAYER3$  then
8:     if  $count\_next = true$  then
9:        $data\_addr[0] \leftarrow count$ 
10:       $data\_addr[1] \leftarrow count$ 
11:     else
12:        $data\_addr[0] \leftarrow data\_addr\_input$ 
13:        $data\_addr[1] \leftarrow data\_addr\_input$ 
14:     end if
15:   end if
16:   Module 2: Counter Update
17:   if  $start = 1$  and  $state\_reg = ST\_CAL$  then
18:      $count \leftarrow count + stride$  ▷Stride-controlled counter
     increment
19:   end if
20:   Module 3: Counter Update Decision
21:   if  $state\_reg = ST\_LAYER1 \vee state\_reg =$ 
      $ST\_LAYER2 \vee state\_reg = ST\_LAYER3$  then
22:      $count\_next \leftarrow (image\_index > 8)$ 
23:   else if  $state\_reg = ST\_SEND$  then
24:      $count\_next \leftarrow send\_cnt$ 
25:   else
26:      $count\_next \leftarrow false$ 
27:   end if
28:   Module 4: Index Update
29:    $data\_addr\_input \leftarrow count +$ 
      $\{4'b0, image\_index\}$  ▷Zero-extended index offset
30:   if  $count\_next = true$  or  $state\_reg \neq state\_next$ 
     then
31:     if  $state\_reg = ST\_LAYER1$  then
32:        $image\_index \leftarrow 7$ 
33:     else if  $state\_reg = ST\_LAYER2 \vee state\_reg =$ 
        $ST\_LAYER3$  then
34:        $image\_index \leftarrow 8$ 
35:     else
36:        $image\_index \leftarrow 0$ 
37:     end if
38:   else
39:      $image\_index \leftarrow image\_index + 1$ 
40:   end if
41:   Output:  $data\_addr[0], data\_addr[1]$ 
42: end for

```

ducibility, detailed FPGA resource utilization and hardware evaluation settings are further reported in Section V.

TABLE I
COMPARISON OF FPGA-BASED SNN ACCELERATORS IN TERMS OF PLATFORM, PERFORMANCE, AND ENERGY EFFICIENCY

	<i>This Work</i>	TCAS-I [19]	Cerebrion [20]	SyneNN [21]	E3NE [22]	FireFly [23]	STI-SNN [24]
FPGA Platform	zcu106	Virtex 7	xc7z100	xczu9eg	xcvu13p	xczu3eg	zcu102
Weight Precision	INT 8	Float 16	INT 8	INT 8	INT 6	INT 8	INT 8
Neuron Model	LIF	LIF	LIF	LIF	LIF	LIF	LIF
Support Network	Spiking Transformer	SFNN	SCNN	SCNN	SCNN	SCNN	SCNN
Resource Utilization LUTs DSP BRAM	225k 120 424	485k 2800 -	86k 0 283	213.8k 681 -	- - -	42k 1152 25	25.52k - 527.5
Power (W)	0.615	1.6	1.4	0.4	2.55	3.6	1.53
Throughput (GOP/s)	79.7	-	44.2	63.62	91	274.49	20.6
Peak Efficiency (GOPs/W)	259.2	0.04	31.57	159.04	36.5	107.64	13.46

D. Low-Power Design

The proposed architecture is specifically designed for implantable visual prosthesis systems, where low and stable power consumption is a critical requirement. The low-power characteristic of the proposed processor originates from three aspects. First, DSC is employed in retinal cell modules to reduce computational complexity. Compared with standard convolution, the MAC operations are reduced from $HWCinCoutK^2$ to $HWCinK^2 + HWCinCout$, which significantly decreases arithmetic operations and SRAM accesses. For a typical 3×3 convolution with $Cout = 64$, DSC reduces approximately 87 % of the MAC operations, thereby lowering hardware switching activity and dynamic power consumption. Second, the proposed multiside sparsity exploitation strategy leverages spike sparsity, weight sparsity, and matrix-structure sparsity to skip redundant computations. Since only active spikes trigger operations, the switching activity is significantly reduced, leading to lower dynamic power. Finally, INT8 quantization is adopted throughout the architecture to reduce multiplier complexity, storage requirements, and memory bandwidth while maintaining high biological prediction accuracy.

V. EVALUATION

A. Experiment Settings

1) *Algorithm Setup*: We conducted experiments using a publicly available macaque V1 dataset acquired with 32-channel microelectrode arrays implanted in the primary V1 of macaques. The dataset contains 7250 distinct visual stimuli and neuronal responses from 166 V1 neurons. Each stimulus was presented four times along four diagonal motion trajectories, each oriented at 45° with respect to the image axes and lasting 250 ms. Accordingly, each stimulus is associated with four trial-specific V1 response recordings. The displayed visual sequences were captured using an iniVation DAVIS346 DVS, which provides a spatial resolution of 346×260 pixels and a temporal resolution of $1 \mu s$. The sensor outputs asynchronous events in an address-event representation, where each event is described by its timestamp, pixel coordinates, and polarity. The resulting continuous spike sequences were used as model inputs, while the recorded V1 neuronal responses served as target outputs for supervised learning. To prevent information leakage, we randomly split the dataset at the level of distinct visual stimuli, with 80 % of the stimuli were used

for training and the remaining 20 % for evaluation. All four presentations and their corresponding V1 response recordings for the same stimulus were assigned to the same subset. Model training was performed for 70 epochs using the Adam optimizer, with an initial learning rate of 1×10^{-3} and an L_2 regularization (weight decay) coefficient of 1×10^{-3} . A MultiStepLR learning-rate scheduler was applied to improve convergence and mitigate overfitting, reducing the learning rate by a factor of 0.1 at epochs 20 and 50. The batch size was set to 25 spike sequences, and mean squared error (mse) was adopted as the optimization objective. To improve training stability, the internal neuronal states were reset after each iteration to prevent undesired temporal accumulation across batches.

2) *Hardware Parameters*: The proposed hardware includes 8 KB of SRAM and 166 neurons. We verify the functional correctness of the hardware through RTL-level implementation. The ZCU106 FPGA platform is used for RTL verification, throughput measurement, and resource utilization analysis. The ZCU106 platform features 1728 DSP units, 469.8 k flip-flops (FFs), and 11 Mb BRAM modules. To better evaluate implantable deployment feasibility, we additionally synthesize the RTL design using Synopsys Design Compiler under the TSMC 40 nm CMOS process. SRAM energy consumption and area are estimated using the CACTI tool [25]. On the contrary, the FPGA evaluation is mainly intended for functional verification and architecture-level throughput analysis. The FPGA-measured power consumption of 0.615 W reported in Table I includes board-level on-chip dynamic power estimated by Vivado power analyzer under the ZCU106 platform. Due to differences in FPGA fabric overhead, routing resources, clock networks, process technology, and measurement methodology, FPGA power numbers are not directly equivalent to ASIC postsynthesis estimations. For comparative experiments, we selected six representative FPGA-based SNN acceleration works. Since the compared works adopt different FPGA devices, network structures, neuron models, operating frequencies, and benchmark tasks, the comparison mainly reflects relative architectural efficiency trends rather than strictly matched implementation conditions.

B. Experimental Results

The similarity in the spatiotemporal domain represents how the predicted values from multiple neurons for various scenes align with the spike counts recorded from V1 cells.

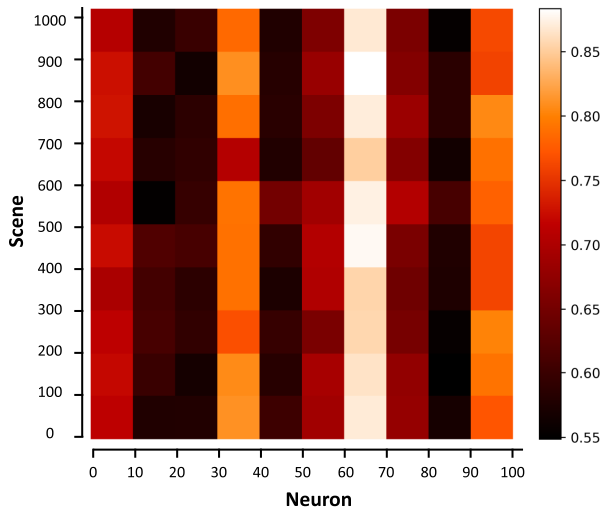


Fig. 7. Matrix of Pearson correlation coefficients in the local spatiotemporal domain. Each small grid represents the Pearson correlation coefficient between the predicted results of ten neurons for 100 natural scenes and the corresponding recorded values.

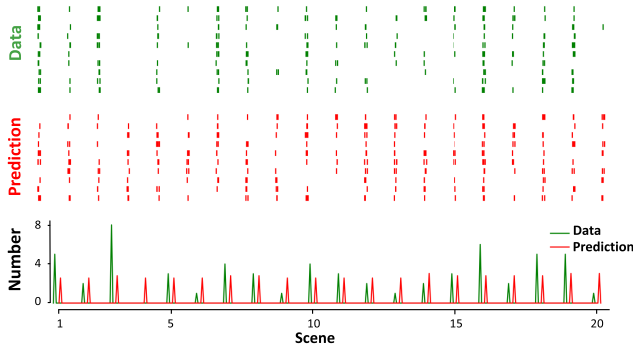


Fig. 8. Comparison of data and model discharge trains across 20 scenes. The data represent one of the four recorded V1 responses, where the V1 response is the firing rate of a single neuron in response to various scene stimuli, expressed in the time domain. Prediction refers to the firing rates predicted by the proposed model for different scene stimuli. The discharge plots below show the firing rates. The spike trains are generated based on the firing rates of a single neuron in response to a specific scene stimulus, with the generation process following a Poisson distribution. This generation process is repeated ten times to mitigate randomness, and the discharge sequences for other scenes are generated in the same manner.

Performance in the spatiotemporal domain is evaluated using error analysis and the Pearson correlation coefficient. Fig. 7 presents the matrix of Pearson correlation coefficients in the local spatiotemporal domain. Fig. 8 shows the distribution of differences between the predicted results and recorded values. Furthermore, to evaluate the impact of low-bit quantization on biological fidelity, we compared the prediction accuracy of the proposed framework under full-precision and 8-bit quantized implementations. Biological fidelity was assessed using the correlation coefficient between the predicted neural responses and the recorded ground-truth signals. The full-precision model achieved a correlation coefficient of 0.73, whereas the 8-bit quantized model achieved a slightly lower value of 0.713. This marginal difference indicates that 8-bit quantization preserves most of the biologically relevant response characteristics while significantly reducing numerical precision. These results suggest that the proposed

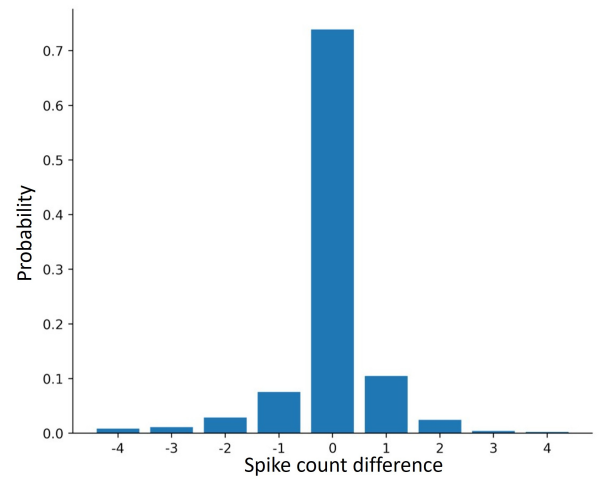


Fig. 9. Spike count difference between recording and predicted value. Histogram showing the prediction error between the model's spike output and experimental V1 recordings across various visual scenes. The majority of differences fall within narrow intervals, indicating high consistency between predicted and actual responses.

neuromorphic framework is robust to moderate quantization and supports efficient hardware deployment without substantial loss of biological plausibility. The results outperform previous state-of-the-art works [4], [12]. These results suggest that in the spatiotemporal domain, the predicted responses are well-matched to the V1 responses. To examine the role of SSA, we conducted an ablation study under the same training and evaluation settings. Removing SSA reduces the Pearson correlation from 0.713 to 0.679 under 8-bit quantization, corresponding to an absolute decrease of 0.034 and a relative decrease of approximately 4.8 %. This result demonstrates that SSA provides a substantive contribution to V1 response prediction under low-bit deployment. In addition to its biologically motivated role in modeling bottom-up visual modulation, SSA helps preserve prediction accuracy after quantization. We evaluated the model's prediction quality using both error statistics and correlation metrics. The histogram of prediction errors is shown in Fig. 9. Prediction error is defined as the predicted spike count minus the recorded spike count. The intervals $[-0.5, 0.5]$ and $[0.5, 1.5]$ contain 75 % and 12 % of the total samples, respectively, accounting for 87 % in total. These statistics indicate that the predicted spike counts are concentrated within narrow error ranges relative to the recorded V1 responses.

The proposed processor is synthesized under the TSMC 40-nm CMOS process at an operating frequency of 500 kHz. Postsynthesis estimation shows that the design achieves a power consumption of 199.10 mW with a chip area of 7.89 mm², which satisfies the 538-mW power requirement of an implantable device. To further evaluate the robustness of the architecture for implantable visual prosthesis applications, we simulated power behavior under varying visual complexities. Results in Fig. 10 indicate that the power consumption exhibits an approximately linear relationship with input toggle rate, while the overall fluctuation remains below 0.01 mW. Such stable power behavior is particularly important for long-term implantable systems that require predictable thermal and

TABLE II
FPGA RESOURCE UTILIZATION

	LUTs	FFs	BRAM	DSP
Top	225k	125k	424	120
Retina Cell	65k	23k	390	120
Attention	70k	56k	1	0
V1 Readout	77k	43k	1	0

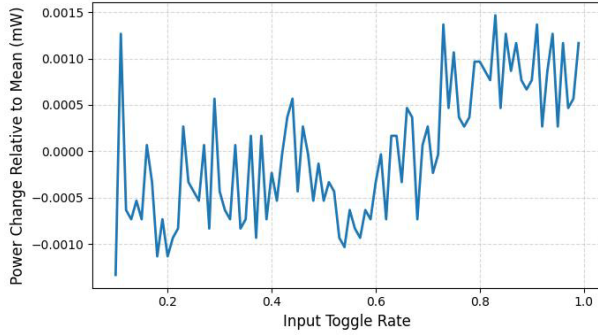


Fig. 10. Relationship between power change relative to mean and input toggle rate. The mean power consumption is 199.11 mW. The higher the input toggle rate, the more complex the visual environment. The power remains remarkably stable within a narrow range, indicating that the system exhibits weak dependence on input complexity and is well-suited for long-term implantable applications.

energy characteristics. To enable comparison with existing FPGA-based SNN accelerators, we additionally implemented the proposed architecture on the Xilinx ZCU106 FPGA platform and evaluated its throughput, power consumption, and energy efficiency under FPGA deployment conditions. The FPGA implementation achieves a throughput of 79.7 GOP/s with measured board-level power consumption of 0.615 W. The comparison results with representative FPGA-based SNN accelerators are summarized in Table I, while detailed FPGA resource utilization is reported in Table II. It should be noted that the compared works differ in multiple aspects, including FPGA platforms, operating frequencies, neuron models, supported network types, numerical precision, and application scenarios. Therefore, the comparison is intended to provide a relative indication of architectural efficiency rather than a strictly controlled benchmark comparison. Despite these differences, the proposed architecture demonstrates competitive energy efficiency advantages. Compared with prior works, our design extends support from conventional SCNN/SFNN models to spike-transformer-based visual prosthesis workloads while maintaining low power consumption. In particular, the proposed design achieves a peak efficiency of 259.2 GOPS/W, outperforming TCAS-I [19], Cerebrion [20], SyncNN [21], and STI-SNN [24]. Although E3NE [22] and FireFly [23] achieve relatively high throughput, their power consumption reaches 2.55 and 3.6 W, respectively, leading to lower overall energy efficiency compared with the proposed architecture.

To enable a better comparison with other works, we implemented and evaluated its performance on the Xilinx ZCU106 FPGA platform and conducted a systematic comparison with existing representative designs. The main characteristics and

performance metrics of each method are summarized in Table I. The detailed resource utilization is presented in Table II. Similar to other designs, our work adopts Int-8 weight precision and the LIF neuron model, but extends the supported network types to spiking Transformers, whereas existing designs are mostly focused on SFNN or SCNN architectures. In terms of experimental results, the proposed design achieves a throughput of 79.7 GOP/s with a power consumption of only 0.615 W, which is higher than Cerebrion [20], SyncNN [21], and STI-SNN [24]. Regarding peak energy efficiency, our design reaches 259.2 GOPS/W, which is significantly higher than existing works: 6480 $\times$ that of TCAS-I [19], 8 $\times$ that of Cerebrion [20], 1.6 $\times$ that of SyncNN [21], and 19.2 $\times$ that of STI-SNN [24]. It is noteworthy that, although E3NE and FireFly exhibit relatively high throughput, their power consumption reaches 2.55 and 3.6 W, respectively, resulting in energy efficiencies of only 36.5 and 107.64 GOPS/W, both still lower than our work [22], [23]. The proposed architecture achieves industry-leading energy efficiency by significantly reducing power consumption while maintaining high throughput. This is attributed to our optimization strategies for spiking Transformers, including parallel structure design both intermodule and intramodule, as well as the full exploitation of multidimensional sparsity.

C. Discussion

Macaque V1 has been extensively shown to share a high degree of structural and functional homology with the human primary V1. This similarity includes comparable receptive field properties, such as orientation selectivity and spatial frequency tuning, as well as columnar organization. As a result, macaque V1 neural responses are widely adopted as a biologically plausible proxy for studying early stage human visual processing. They are also commonly used to validate vision models intended for translational applications.

Although our quantitative evaluation is conducted using macaque V1 data, the proposed framework is not intrinsically species specific. Instead, it is formulated as a parameter-tunable neuromorphic modeling system. The framework abstracts early visual encoding through biologically interpretable components, including spiking-based retinal encoding, synaptic weight modulation, and temporal integration mechanisms. These parameters can be systematically adjusted to account for interspecific differences in cortical response statistics. They can also capture interindividual variability observed in the human V1.

Moreover, the framework is inherently adaptable to diverse patient populations with different forms and severities of retinal degeneration. Various pathological conditions, such as

photoreceptor loss, altered bipolar cell signaling, or abnormal ganglion cell firing patterns, can be modeled within the same architecture. This is achieved by modifying retinal encoding parameters, spike sparsity constraints, and synaptic weighting schemes, without changing the overall system design. Such flexibility enables the framework to emulate a broad range of degeneration-induced neural response patterns. It also supports personalized parameter calibration when patient-specific neural or behavioral data are available.

We acknowledge that direct validation using human cortical recordings and clinical datasets is an important direction for future work. Nevertheless, the current macaque-based evaluation provides a biologically grounded and experimentally controllable testbed. It demonstrates the general modeling capability and adaptability of the proposed framework. Importantly, the framework is designed to generalize across species and pathological conditions, rather than to fit a single dataset. From an instrumentation and measurement perspective, the proposed framework provides an event-driven pipeline for visual-stimulus sensing, spike-domain processing, and quantitative evaluation of predicted cortical responses. The Pearson correlation analysis, spike-count error distribution, input-toggle-rate power analysis, and FPGA resource-utilization results jointly characterize prediction fidelity and hardware behavior under implantable-system constraints.

VI. CONCLUSION

In this work, we propose a vision prosthesis with a codesign of hardware and software. Inspired by the human retina and visual pathways, we design a three-layer spiking convolutional module to simulate the photoreceptors, bipolar cells, and ganglion cells of the retina. In addition, we incorporate a spiking attention mechanism to model the bottom-up attention pathway. Finally, a spiking linear layer predicts the V1 neuronal responses. Compared to real responses, the model achieves a Pearson correlation coefficient of 0.713, surpassing existing methods. Based on the proposed model, we develop a specialized hardware architecture that leverages pipelined parallel processing within and across modules, as well as multidimensional sparsity exploration, to enhance performance. The energy efficiency of this design exceeds that of state-of-the-art neuromorphic chips.

REFERENCES

- [1] H. Wang et al., "OCTFormer: An efficient hierarchical transformer network specialized for retinal optical coherence tomography image recognition," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–17, 2023.
- [2] W. G. Chung et al., "Liquid-metal-based three-dimensional microelectrode arrays integrated with implantable ultrathin retinal prosthesis for vision restoration," *Nature Nanotechnol.*, vol. 19, no. 5, pp. 688–697, May 2024.
- [3] Q. Yan et al., "Revealing fine structures of the retinal receptive field by deep-learning networks," *IEEE Trans. Cybern.*, vol. 52, no. 1, pp. 39–50, Jan. 2022.
- [4] G. Pogoncheff, J. Granley, and M. Beyeler, "Explaining V1 properties with a biologically constrained deep learning architecture," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 13908–13930.
- [5] S. Idrees, M. B. Manookin, F. Rieke, G. D. Field, and J. Zylberberg, "Biophysical neural adaptation mechanisms enable artificial neural networks to capture dynamic retinal computation," *Nature Commun.*, vol. 15, no. 1, p. 5957, Jul. 2024.
- [6] M. Cui, Y. Zhu, Y. Liu, Y. Liu, G. Chen, and K. Huang, "Dense depth-map estimation based on fusion of event camera and sparse LiDAR," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.
- [7] D. D. Zhou, J. D. Dorn, and R. J. Greenberg, "The Argus II retinal prosthesis system: An overview," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, Jul. 2013, pp. 1–6.
- [8] K. Stingl et al., "Interim results of a multicenter trial with the new electronic subretinal implant alpha AMS in 15 patients blind from inherited retinal degenerations," *Frontiers Neurosci.*, vol. 11, p. 445, Aug. 2017.
- [9] L. McIntosh, N. Maheswaranathan, A. Nayeibi, S. Ganguli, and S. Baccus, "Deep learning models of the retinal response to natural scenes," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.
- [10] E. Ho, A. Shmakov, and D. Palanker, "Decoding network-mediated retinal response to electrical stimulation: Implications for fidelity of prosthetic vision," *J. Neural Eng.*, vol. 17, no. 6, Dec. 2020, Art. no. 066018.
- [11] J. A. Leñero-Bardallo, T. Serrano-Gotarredona, and B. Linares-Barranco, "A 3.6 μ s latency asynchronous frame-free event-driven dynamic-vision-sensor," *IEEE J. Solid-State Circuits*, vol. 46, no. 6, pp. 1443–1455, Jun. 2011.
- [12] C. Wang, J. Yang, and M. Sawan, "NeuroSEE: A neuromorphic energy-efficient processing framework for visual prostheses," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 8, pp. 4132–4141, Aug. 2022.
- [13] F. Akopyan et al., "TrueNorth: Design and tool flow of a 65 mW 1 million neuron programmable neurosynaptic chip," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 34, no. 10, pp. 1537–1557, Oct. 2015.
- [14] M. Davies et al., "Loihi: A neuromorphic manycore processor with on-chip learning," *IEEE Micro*, vol. 38, no. 1, pp. 82–99, Jan. 2018.
- [15] S. Narayanan, K. Taht, R. Balasubramanian, E. Giacomini, and P.-E. Gaillardon, "SpinalFlow: An architecture and dataflow tailored for spiking neural networks," in *Proc. ACM/IEEE 47th Annu. Int. Symp. Comput. Archit. (ISCA)*, May 2020, pp. 349–362.
- [16] F. Liu et al., "SATO: Spiking neural network acceleration via temporal-oriented dataflow and architecture," in *Proc. 59th ACM/IEEE Design Autom. Conf.*, Jul. 2022, pp. 1105–1110.
- [17] S. Kim, S. Kim, W. Jo, S. Kim, S. Hong, and H.-J. Yoo, "20.5 C-transformer: A 2.6–18.1 μ J/token homogeneous DNN-transformer/spiking-transformer processor with big-little network and implicit weight generation for large language models," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 368–370.
- [18] C. Fang et al., "A 0.078 pJ/SOP unstructured sparsity-aware spiking attention/convolution processor with 3D compute array," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Apr. 2024, pp. 1–2.
- [19] S. Li, Z. Zhang, R. Mao, J. Xiao, L. Chang, and J. Zhou, "A fast and energy-efficient SNN processor with adaptive clock/event-driven computation scheme and online learning," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 4, pp. 1543–1552, Apr. 2021.
- [20] Q. Chen, C. Gao, and Y. Fu, "Cerebron: A reconfigurable architecture for spatiotemporal sparse spiking neural networks," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 30, no. 10, pp. 1425–1437, Oct. 2022.
- [21] S. Panchapakesan, Z. Fang, and J. Li, "SyncNN: Evaluating and accelerating spiking neural networks on FPGAs," *ACM Trans. Reconfigurable Technol. Syst.*, vol. 15, no. 4, pp. 1–27, Dec. 2022.
- [22] D. Gerlinghoff, Z. Wang, X. Gu, R. S. M. Goh, and T. Luo, "E3NE: An end-to-end framework for accelerating spiking neural networks with emerging neural encoding on FPGAs," *IEEE Trans. Parallel Distrib. Syst.*, vol. 33, no. 11, pp. 3207–3219, Nov. 2022.
- [23] J. Li, G. Shen, D. Zhao, Q. Zhang, and Y. Zeng, "FireFly: A high-throughput hardware accelerator for spiking neural networks with efficient DSP and memory optimization," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 31, no. 8, pp. 1178–1191, Aug. 2023.
- [24] K. Wang, C. Yang, C. Yu, Y. Sin Ang, B. Wang, and A. Wang, "STI-SNN: A 0.14 GOPS/W/PE single-timestep inference FPGA-based SNN accelerator with algorithm and hardware co-design," 2025, *arXiv:2506.08842*.
- [25] S. Thoziyoor, N. Muralimanohar, J. H. Ahn, and N. P. Jouppi, "CACTI 5.1," HP Labs, Palo Alto, CA, USA, Tech. Rep. HPL-2008-20, 2008.